

Advanced Micrograph Chemometrics Handout: Bridging Digital Image Matrices and Material Surface Physics

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1 Introduction: Beyond Manual Feature Measuring

In advanced materials characterization (e.g., analyzing Covalent Organic Frameworks (COFs), graphene boundaries, or porous membranes), evaluating structural morphology from micrographs using manual point-and-click tools is mathematically insufficient. Manual profiling introduces observer bias, misses complex spatial networks, and lacks the statistical power required to track subtle variations across different synthesis runs.

The computational pipeline implemented in `module4-SEM-grains-advanced` transforms 2D image brightness arrays into deep, quantitative descriptors. This manual maps the code’s underlying matrix mathematics onto real physical phenomena, separating standard shape analysis from non-standard, research-grade topology and surface texture entropy.

2 Tier 1: Preprocessing & Structural Segmentation

2.1 1. Noise Filtering via Grayscale Medians

Microscopic imaging systems inherently introduce high-frequency detector electronic noise and thermal scattering variations. Prior to segmentation, the pipeline uses a **Median Filter** over an operational spatial coordinate disk ($r = 2$ px):

```
1 denoised = filters.median(norm_img, morphology.disk(2))
```

Unlike standard Gaussian blurring, which averages neighboring edge values and smears crisp structural transitions, a median filter replaces each pixel with the exact median value of its localized neighbors. This preserves sharp solid-to-void interfaces while eliminating single-pixel detector artifacts.

2.2 2. Local Adaptive Binarization

Standard global thresholding methods assume that background intensity remains completely uniform across the entire image frame. However, experimental micrographs regularly feature large-scale lighting gradients caused by sample tilt, uneven electron beam dispersion, or surface charging shadows.

To overcome this, the code uses an adaptive binarization threshold:

```
1 thresh_local = filters.threshold_local(denoised, block_size=45, offset
    =0.01)
2 binary_mask = denoised > thresh_local
```

Mathematically, a unique threshold $T(x, y)$ is dynamically generated for every individual pixel coordinates based on the local mean intensity within an surrounding window (`block_size` = 45 px):

$$T(x, y) = \mu_{\text{local}}(x, y) - \text{offset} \quad (1)$$

The `block_size` must always be chosen as an odd integer slightly larger than the average structural domain feature size to prevent the algorithm from interpreting the interior of a grain as background noise.

2.3 3. Marker-Controlled Watershed Allocation

When individual grain boundaries grow tight against each other, standard contour-finding tools merge them into a single massive cluster, distorting the true feature counts. The code resolves these connected areas using a marker-controlled watershed segmentation.

First, the binary mask is converted into a topographical distance map via the Euclidean Distance Transform (EDT):

$$D(x, y) = \min_{(x_b, y_b) \in \text{Background}} \sqrt{(x - x_b)^2 + (y - y_b)^2} \quad (2)$$

The core geometric centers of tightly packed grains appear as local peaks in this distance matrix. These peak coordinates are isolated using `feature.peak_local_max(..., min_distance=12)` to act as unambiguous starting markers. The pipeline then mathematically inverts the map $(-D(x, y))$ and “floods” it with water starting from each marker point. Where independent rising fluid fronts meet, an exact, sub-pixel dividing boundary is drawn, cleanly separating touching particles.

3 Tier 2: Standard Geometrical Metrics

Commercial microscope software often relies on basic geometric values derived from the separated object labels. The pipeline calculates these standard parameters automatically using `measure.regionprops_table`:

3.1 1. Domain Circularity Factor

Tracks the perimeter-to-area relationship of each detected domain to evaluate its circular shape irregularity:

$$\text{Circularity} = \frac{4\pi \times \text{Area}}{\text{Perimeter}^2} \quad (3)$$

A perfect circle evaluates to 1.0. Lower values quantify irregular, jagged, or elongated crystal growth patterns, providing an indirect indicator of structural deviations from thermodynamic equilibrium.

3.2 2. Object Solidity

Solidity quantifies the density profile of the grain’s perimeter by evaluating the ratio of its true pixel area to its calculated convex hull outline area:

$$\text{Solidity} = \frac{\text{Area}}{\text{Convex Area}} \quad (4)$$

Grains displaying high edge roughness, branching features, or structural facets yield significantly depressed solidity scores, enabling automated screening for highly dendritic crystal growth shapes.

4 Tier 3: Research-Grade Advanced Chemometrics

The core advantage of this advanced script is its ability to unlock deep non-standard analytics that go beyond basic geometry, capturing spatial correlations and fine micro-texture distributions.

4.1 1. Spatial Proximity Topology via Nearest-Neighbor Spacing

Instead of treating each domain like an isolated feature, the pipeline analyzes the spatial arrangement of the particles relative to each other in coordinate space. It extracts the center coordinates $\mathbf{P}_i = (x_i, y_i)$ of each peak and constructs an $N \times N$ spatial distance matrix:

$$d_{ij} = \|\mathbf{P}_i - \mathbf{P}_j\|_2 = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2} \quad (5)$$

By setting the matrix diagonal values to infinity (`np.fill_diagonal(dist_mat, np.inf)`) to ignore each peak's distance to itself, the algorithm identifies the exact straight-line distance to its absolute nearest neighbor:

```
1 nearest_neighbor_distances = np.min(dist_mat, axis=1)
```

The average of these values (μ_{spacing}) yields an empirical estimation of the material's underlying lattice parameter, while the standard deviation (σ_{spacing}) quantifies localized lattice dislocations and amorphous defect densities.

4.2 2. Textural Growth Anisotropy Maps

The orientation angle (θ) represents the tilt of each grain's major elongation axis relative to the horizontal pixel plane. The script plots these orientation variables onto a continuous polar coordinate system:

```
1 ax_orient = fig.add_subplot(grid[1, 1], projection='polar')
```

In image matrix processing, 0° is located at the middle-right (3 o'clock position), and 90° points straight up. If your material grew under a directional force—such as a temperature gradient, a magnetic field, or a liquid shear line—the polar histogram will show a highly localized distribution along that alignment vector rather than a random circle, mapping the sample's structural anisotropy directly.

4.3 3. Surface Texture Entropy (GLCM Analysis)

When the varying grayscale intensities across a material's surface represent real physical topography, the pipeline evaluates its micro-texture using a Grey-Level Co-occurrence Matrix (GLCM).

The GLCM tracks how often a pixel of intensity i occurs at a specific spatial displacement distance ($d = 3$ px) and orientation angle from a pixel of intensity j . From this matrix, the pipeline extracts **Textural Entropy**:

$$\text{Entropy} = - \sum_i \sum_j P(i, j) \log_2 (P(i, j) + \epsilon) \quad (6)$$

Mathematically, entropy measures the state of local structural disorder within the pixel patterns. Samples containing uniform, highly crystalline domains exhibit low texture entropy, whereas amorphous phases or surface defect regions yield elevated entropy values.

5 Instructions for Analyzing Your Group's Micrographs

To run this advanced pipeline on your own experimental samples, transfer your image into the notebook workspace and update the user calibration line at the base of the file:

```
1 # User Configuration Block
2 # Adapt these values to match your custom microscope image attributes:
3 my_micrograph = "image.png" # File name in your environment
4 true_width = 893 # True horizontal pixel dimension
5 true_height = 899 # True vertical pixel dimension
6
7 # Execute the advanced pipeline
8 material_database = advanced_material_micrograph_analytics(
9     filepath=my_micrograph,
10     target_width=true_width,
11     target_height=true_height
12 )
```

5.1 Morphological Optimization Guidelines

Table 1: Adjusting Pipeline Parameters based on Microscopic Artifacts

Observed Microscopic Artifact	Algorithm Failure Mode	Corrective Pipeline Tuning Action
High-frequency detector noise	Over-segmentation / False features	Increase Median Filter disk radius
Large-scale shadows / Illumination tilt	False background boundaries	Decrease the adaptive threshold <code>offset</code>
Faint, low-contrast structural walls	Under-segmentation / Merged labels	Reduce the peak filter <code>min_distance</code>